

ENTHUSIAST, LEADER & ACHIEVER COURSE

PHASE : ALL PHASE

TARGET : PRE MEDICAL 2025

Test Type : **MAJOR**

Test Pattern : **NEET (UG)**

TEST DATE : 17-04-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	4	4	2	1	3	1	1	1	1	1	1	4	3	3	4	4	1	4	2	1	4	2	2	3	4	2	3	3	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	4	4	1	4	3	3	2	2	4	3	4	1	2	4	3	2	4	4	4	1	1	1	2	1	1	2	2	1	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	4	1	2	1	1	2	3	4	1	2	1	1	3	2	3	4	3	1	3	2	4	2	1	2	2	4	2	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	3	3	1	4	3	1	2	2	2	3	3	4	4	2	4	1	1	2	4	2	2	1	1	2	4	4	2	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	2	2	4	1	1	2	2	4	1	1	4	2	4	2	2	3	2	3	2	3	3	2	1	2	3	3	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	1	4	4	4	3	4	2	4	2	2	2	4	3	1	3	1	3	3	3	4	2	4	3	1	2	2	2	1	2

HINT – SHEET

1. Ans (4)

$$\begin{aligned} \text{Area of parallelogram} &= |\bar{a} \times \bar{b}| \\ &= (3\hat{i} + 4\hat{j} + 5\hat{k}) \times (\hat{i} - \hat{j} - \hat{k}) \\ &= |\hat{i} + 8\hat{j} - 7\hat{k}| \\ &= \sqrt{1^2 + 8^2 + (-7)^2} = \sqrt{114} \end{aligned}$$

2. Ans (4)

Angular momentum = $[ML^2T^{-1}]$
 Coeff of thermal conductivity = $[MLT^{-3}K^{-1}]$
 Torque = $[ML^2T^{-2}]$
 Gravitational constant = $[M^{-1}L^3T^{-2}]$
 So, gravitational constant has power of (-1) of M.

3. Ans (4)

Mean of observation

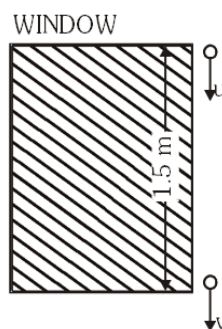
$$\frac{1.25 + 1.24 + 1.27 + 1.21 + 1.28}{5} = 1.25 \text{ s}$$

Mean absolute error

$$= \frac{0 + 0.01 + 0.02 + 0.04 + 0.03}{5} = 0.02 \text{ s}$$

% error = $\frac{0.02}{1.25} \times 100 = 1.6\%$

4. Ans (2)



By using $s = ut + \frac{1}{2}at^2$

$$1.5 = u(0.1) + \frac{1}{2} (10) (0.1)^2$$

$$\Rightarrow 15 = u + 0.5$$

$$\Rightarrow u = 14.5 \text{ ms}^{-1}$$

5. Ans (1)

$$H = \frac{u^2 \sin^2 \theta}{2g} = \frac{(20)^2 \sin^2 30^\circ}{2(10)}$$

$$= 5\text{m}$$

6. Ans (3)

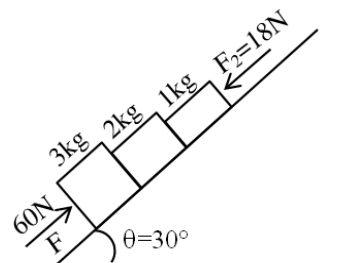
$$\Delta K = \Delta W$$

$$K_f - K_i = \vec{F} \cdot \Delta \vec{x}$$

$$K_f - 10 = -6$$

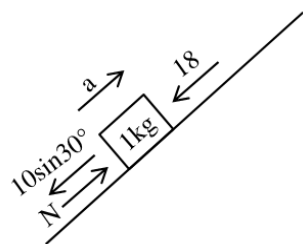
$$K_f = 10 - 6 = 4 \text{ J}$$

7. Ans (1)



$$a = \frac{\text{Net force}}{\text{Total mass}} = \frac{60 - (18 + 60 \sin 30^\circ)}{6} = 2 \text{ ms}^{-2}$$

FBD of 1kg



$$N - 18 - 5 = 1(2)$$

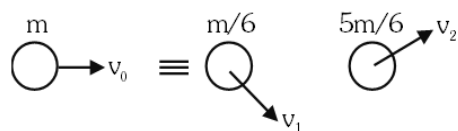
$$N = 25 \text{ N}$$

8. Ans (1)

∴ There is no friction between feet of passenger and floor of bus so passenger will not move back when seen from ground.

So Assertion is false but reason is true.

9. Ans (1)



Conservation of linear momentum,

$$m\vec{v}_0 = \frac{m}{6}\vec{v}_1 + \frac{5m}{6}\vec{v}_2$$

$$\Rightarrow m(20\hat{i} + 25\hat{j} - 12\hat{k}) = \frac{m}{6}(100\hat{i} + 35\hat{j} + 8\hat{k}) + \frac{5m}{6}\vec{v}_2$$

$$\Rightarrow \vec{v}_2 = 4\hat{i} + 23\hat{j} - 16\hat{k}$$

10. Ans (1)

Telegram: @NEETxNOAH

In elastic collision maximum energy is transfer when $M = m$

11. Ans (1)

$$T = m\omega^2 r$$

$$T \propto \omega^2$$

$$\frac{T_2}{T_1} = \left(\frac{\omega_2}{\omega_1}\right)^2$$

$$\Rightarrow \frac{4T}{T} = \left(\frac{\omega_2}{10}\right)^2$$

$$\Rightarrow \omega_2 = 20 \text{ rpm}$$

12. Ans (1)

$$t = I\alpha$$

$$\Rightarrow \alpha = \frac{\tau}{I} = \frac{100}{300} = \frac{1}{3} \text{ rad/sec}^2$$

$$\omega_i = 0$$

$$\omega_f = \omega_i + \alpha t$$

$$= 0 + \frac{1}{3} \times 3$$

$$\omega_f = 1 \text{ rad/sec}$$

13. Ans (4)

$$E_{\text{sphere}} = \frac{1}{2} I_s \omega^2$$

$$= \frac{1}{2} \times \frac{2}{5} MR^2 \times \omega^2$$

$$E_{\text{cylinder}} = \frac{1}{2} I_c (2\omega)^2$$

$$= \frac{1}{2} \times \frac{MR^2}{2} \times 4\omega^2$$

$$\frac{E_{\text{sphere}}}{E_{\text{cylinder}}} = \frac{1}{5}$$

14. Ans (3)

According to question

$$\therefore T^2 \propto R^3$$

$$\therefore \frac{T^2}{x^2} \propto \frac{R^3}{64R^3}$$

$$\Rightarrow x^2 = 64T^2$$

$$\Rightarrow x = 8T$$

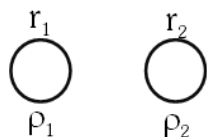
15. **Ans (3)**

From continuity equation,

Speed at Y is less than speed at X

$$\therefore K_2 < K_1$$

16. **Ans (4)**



$$v_T = \frac{2r^2(\sigma - \rho)g}{9\eta}$$

$$\frac{v_1}{v_2} = \left(\frac{r_1}{r_2}\right)^2 \frac{(\sigma_1 - \rho)}{(\sigma_2 - \rho)}$$

$$= \left(\frac{1}{2}\right)^2 \left(\frac{8\rho_2 - 0.1\rho_2}{\rho_2 - 0.1\rho_2}\right) = \frac{79}{36}$$

17. **Ans (4)**

$$W = T(2\Delta A) \quad \{\Delta A = (20 - 8) \text{ cm}^2\}$$

$$\begin{aligned} \Rightarrow T &= \frac{W}{2\Delta A} \\ &= \frac{3 \times 10^{-4}}{2 \times 12 \times 10^{-4}} \\ &= 0.125 \text{ Nm}^{-1} \end{aligned}$$

18. **Ans (1)**

$$\text{Volume of 10g of water} = 10 \text{ cm}^3$$

$$\text{W.D.} = P\Delta V = 10^5 [(11 - 10) \times 10^{-6}] = 0.1 \text{ J}$$

19. **Ans (4)**

By using Wien's law $\lambda_m T = \text{constant}$,

$$\text{we have } T \propto \frac{1}{\lambda_m}$$

$$\text{As } \lambda_B < \lambda_Y < \lambda_R \quad (\text{VIBGYOR})$$

$$\Rightarrow T_A > T_C > T_B$$

20. **Ans (2)**

Telegram: @NEETxNOAH

$$Q = \Delta U + W \Rightarrow mL = \Delta U + P(V_2 - V_1)$$

$$\Rightarrow 1(2256) = \Delta U + 1 \times 10^5 (1670 \times 10^{-6})$$

$$\Rightarrow \Delta U = 2089 \text{ J}$$

21. **Ans (1)**

For S.H.M.

$$\text{Maximum acceleration} = \omega^2 A = \alpha$$

$$\text{Maximum velocity} = \omega A = \beta$$

$$\Rightarrow \omega = \frac{\alpha}{\beta} \Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi\beta}{\alpha}$$

22. **Ans (4)**

For closed pipe frequencies will be,

$$= f_0, 3f_0, 5f_0, 7f_0$$

$$= 85, 255, 425, 595$$

23. **Ans (2)**

$$\begin{aligned} \text{LC} &= \frac{\text{pitch}}{\text{No. of divisions on circular scale}} \\ &= \frac{1}{100} \text{ mm} = 0.01 \text{ mm} \end{aligned}$$

$$\text{Reading} = \text{MSR} + \text{LC} \times \text{CSR}$$

$$= 1 + 0.01 \times 63$$

$$= 1.63 \text{ mm} = 0.163 \text{ cm}$$

24. **Ans (2)**

$$\frac{n_1}{n_2} = \frac{\frac{v_1}{4\ell}}{\frac{v_2}{4\ell}}$$

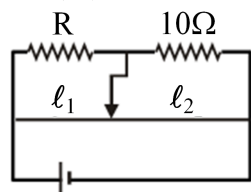
$$\frac{400}{n_2} = \frac{v_1}{v_2} \left[\text{from } v \propto \sqrt{T} \right]$$

$$\frac{400}{n_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\frac{400}{n_2} = \sqrt{\frac{300}{363}}$$

$$\Rightarrow \boxed{n_2 = 440 \text{ Hz}}$$

25. Ans (3)



$$\frac{R}{10} = \frac{l_1}{l_2} \Rightarrow \frac{R}{10} = \frac{3}{2}$$

$$\Rightarrow R = 15\Omega$$

Length of 15Ω resistance wire is 1.5 m

$$\therefore \text{length of } 1\Omega \text{ resistance wire} = \frac{1.5}{15} = 0.1 \text{ m}$$

26. Ans (4)

Shell is equipotential surface

$$\text{So, } V_p = V_c$$

$$\therefore \text{Potential Difference} = 0$$

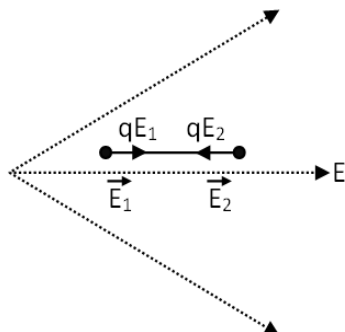
27. Ans (2)

$$|\vec{E}_1| > |\vec{E}_2|$$

as field lines are closer at charge +q,

So, net force on the dipole acts towards right side.

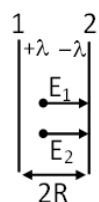
Also, A system always moves to decrease its potential energy.



28. Ans (3)

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$E = E_1 + E_2$$

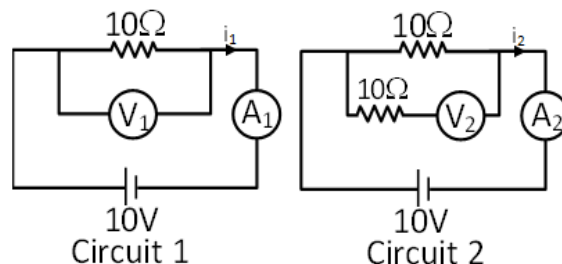


$$E = \frac{\lambda}{2\pi\epsilon_0 R} + \frac{\lambda}{2\pi\epsilon_0 R}$$

$$E = \frac{\lambda}{\pi\epsilon_0 R} \text{ N/C}$$

29. Ans (3)

Telegram: @NEETxNOAH



10Ω is in series with ideal voltmeter. Therefore it will not affect the circuit (Circuit-2)

$$i_1 = \frac{10}{10} = 1\text{A}$$

$$i_2 = \frac{10}{10} = 1\text{A}$$

$$V_1 = 10\text{V}$$

$$V_2 = 10\text{V}$$

30. Ans (1)

$$(A) v_d = \left(\frac{eE}{m} \right) \tau$$

$$(B) J = \sigma E = E/\rho \Rightarrow \rho = E/J$$

$$(C) \rho = \frac{E}{nev_d}$$

$$v_d = \frac{E}{ne\rho}$$

$$\frac{eE}{m} \tau = \frac{E}{ne\rho}$$

$$\tau = \frac{m}{ne^2\rho}$$

$$(D) i = neAv_d$$

$$\frac{i}{A} = nev_d$$

$$J = nev_d$$

31. Ans (3)

$$\mu = \frac{v_d}{E} = \frac{7.5 \times 10^{-4}}{3 \times 10^{-10}} = 2.5 \times 10^6$$

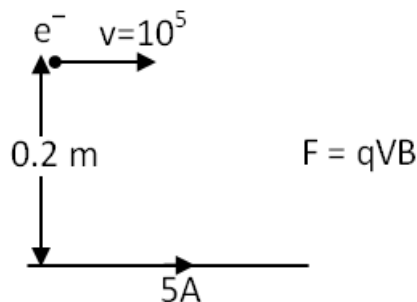
32. Ans (4)

$$\text{In parallel } \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{K_{eq}(2A)}{\ell} = \frac{K_1 A}{\ell} + \frac{K_2 A}{\ell}$$

$$K_{eq} = \frac{K_1 + K_2}{2}$$

33. Ans (4)

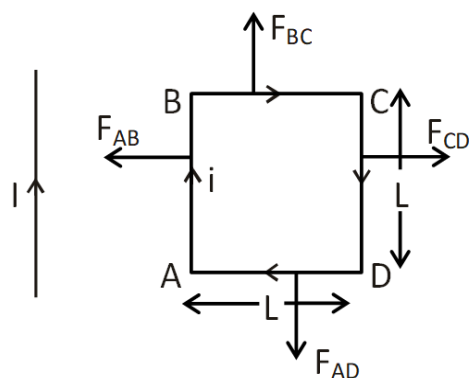


$$F = ev \left(\frac{\mu_0 i}{2\pi r} \right)$$

$$F = \frac{1.6 \times 10^{-19} \times 10^5 \times 2 \times 10^{-7} \times 5}{0.2}$$

$$F = 8 \times 10^{-20} \text{ Newton}$$

34. Ans (1)



$$F_{AB} = i \ell B \text{ (Attractive)}$$

$$F_{AB} = i(L) \cdot \frac{\mu_0 I}{2\pi \left(\frac{L}{2} \right)} (\leftarrow) = \frac{\mu_0 i I}{\pi} (\leftarrow)$$

$$F_{BC} (\uparrow) \text{ and } F_{AD} (\downarrow) \Rightarrow \text{cancels each other}$$

$$F_{CD} = i \ell B \text{ (Repulsive)}$$

$$F_{CD} = i(L) \frac{\mu_0 I}{2\pi \left(\frac{3L}{2} \right)} (\rightarrow) = \frac{\mu_0 i I}{3\pi} (\rightarrow)$$

$$\Rightarrow F_{\text{net}} = \frac{\mu_0 i I}{\pi} - \frac{\mu_0 i I}{3\pi} = \frac{2\mu_0 i I}{3\pi}$$

35. Ans (4)

Telegram: @NEETxNOAH

$$\text{Flux linked with each turn} = 4 \times 10^{-3} \text{ Wb}$$

$$\therefore \text{Total flux linked} = 1000[4 \times 10^{-3}] \text{ Wb} = 4 \text{ Wb}$$

$$\phi_{\text{total}} = L i = 4 \Rightarrow L = 1 \text{ H}$$

36. Ans (3)

$$Q = \frac{\omega}{\Delta\omega} = \frac{\omega L}{R}$$

$$\Rightarrow \Delta\omega = R/L = \frac{50}{4} = 8 \text{ rad/sec}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{5 \times 80 \times 10^{-6}}} = 50 \text{ rad/sec.}$$

$$\omega_{\text{min}} = \omega_0 - \frac{\Delta\omega}{2} = 46 \text{ rad/sec}$$

$$\omega_{\text{max}} = \omega_0 + \frac{\Delta\omega}{2} = 54 \text{ rad/sec}$$

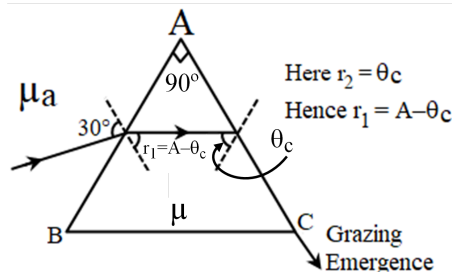
37. Ans (3)

$$n = \sqrt{\epsilon_r \mu_r}$$

$$n = \frac{c}{v} \Rightarrow v = \frac{c}{n}$$

$$v = \left(\frac{c}{\sqrt{\epsilon_r \mu_r}} \right)$$

38. Ans (2)



By snell's law,

$$\mu_a \sin 30^\circ = \mu \sin r_1$$

$$\Rightarrow (1) \sin 30^\circ = \mu \sin(A - \theta_c)$$

$$\Rightarrow \sin 30^\circ = \mu \sin(90^\circ - \theta_c)$$

$$\Rightarrow \sin 30^\circ = \mu \cos \theta_c$$

$$\Rightarrow \sin 30^\circ = \mu \frac{\sqrt{\mu^2 - 1}}{\mu}$$

$$\therefore \left(\cos \theta_c = \frac{\sqrt{\mu^2 - 1}}{\mu} \right)$$

$$\Rightarrow \frac{1}{2} = \sqrt{\mu^2 - 1} \Rightarrow \frac{1}{4} = \mu^2 - 1$$

$$\Rightarrow \mu^2 = 5/4 \Rightarrow \mu = \frac{\sqrt{5}}{2}$$

39. Ans (2)

$$\frac{1}{F_1} = \frac{1}{f} + \frac{1}{f}$$

$$\Rightarrow F_1 = f/2$$

$$\text{and } F_2 = f$$

$$\Rightarrow \frac{F_1}{F_2} = \frac{1}{2}$$

40. Ans (4)

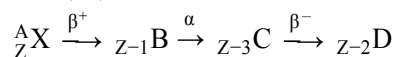
$$\text{For first minima, } \sin 30^\circ = \frac{\lambda}{a} = \frac{1}{2}$$

First secondary maxima will be at

$$\sin \theta = \frac{3\lambda}{2a} = \frac{3}{2} \left(\frac{1}{2} \right)$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{3}{4} \right)$$

41. Ans (3)



β^+ decreases atomic number by 1

α decreases atomic number by 2

β^- increases atomic number by 1

42. Ans (4)

$$I = \frac{E}{At}$$

$$\Rightarrow E = IAt = \frac{20}{10^{-4}} \times 20 \times 10^{-4} \times 60$$

$$= 24 \times 10^3 \text{ J}$$

43. Ans (1)

$$\lambda = 1.227 \times 10^{-2} \text{ nm} = 0.1227 \text{ \AA}$$

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

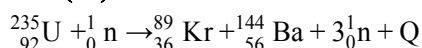
$$\Rightarrow 0.1227 = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

$$\Rightarrow \sqrt{V} = 10^2$$

$$\Rightarrow V = 10^4 \text{ volt}$$

44. Ans (2)

Telegram: @NEETxNOAH



45. Ans (4)

$$Y = (A + B)C = 1 \Rightarrow C = 1$$

46. Ans (3)

For CH_3COOH :

$$[\text{H}^+] = C \cdot \alpha$$

$$[\text{H}^+] = 10^{-2} \times \frac{1}{100} = 10^{-4} \text{ M}$$

$$\text{pH} = -\log[\text{H}^+]$$

$$\text{pH} = -\log 10^{-4}$$

$$\text{pH} = 4$$

47. Ans (2)

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

$$\frac{0.693}{2000} = \frac{2.303}{8000} \log \frac{[A]_0}{0.02}$$

$$\therefore [A]_0 = 0.32 \text{ M}$$

48. Ans (4)

Relative lowering in vapour pressure = Mole fraction

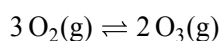
of solute

$$\therefore x_{\text{solute}} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}}$$

$$= \frac{1}{1 + 3}$$

$$= \frac{1}{4} = 0.25$$

49. Ans (4)



$$K_c = \frac{[\text{O}_3]^2}{[\text{O}_2]^3}$$

$$3 \times 10^{-59} = \frac{[\text{O}_3]^2}{(4 \times 10^{-2})^3}$$

$$[\text{O}_3]^2 = 3 \times 10^{-59} \times 64 \times 10^{-6}$$

$$[\text{O}_3]^2 = 19.2 \times 10^{-64}$$

$$[\text{O}_3] = 4.38 \times 10^{-32} \text{ M}$$

50. Ans (4)

Acc. to Bohr's atomic model

$$r \propto \frac{n^2}{Z} \quad 3^{\text{rd}} \text{ orbit of } \text{Li}^{+2} \quad n_1 = 3$$

$$Z_1 = 3$$

$$\Rightarrow \quad 2^{\text{nd}} \text{ orbit of } \text{He}^+ \quad n_2 = 2$$

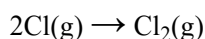
$$Z_2 = 2$$

$$\frac{(r_3)_{\text{Li}^{+2}}}{(r_2)_{\text{He}^+}} = \frac{n_1^2}{n_2^2} \times \frac{Z_2}{Z_1}$$

$$\frac{(r_3)_{\text{Li}^{+2}}}{105.8 \text{ pm}} = \frac{3 \times 3}{2 \times 2} \times \frac{2}{3}$$

$$(r_3)_{\text{Li}^{+2}} = 158.7 \text{ pm}$$

51. Ans (1)



$$\Delta_r S < 0 \text{ and } \Delta_r H < 0$$

52. Ans (1)

For reaction $\Delta H = -$ ve and $\Delta n_g = -$ ve

$\therefore$ High P, Low T, favour product formation.

54. Ans (2)

Telegram: @NEETxNOAH

For an ideal solution $\Delta H_{\text{mix}} = 0$

$$\Delta U_{\text{mix}} = 0$$

$$\Delta S_{\text{mix}} \neq 0$$

$$\text{According to } \Delta G_{\text{mix}} = \Delta H_{\text{mix}} - T \Delta S_{\text{mix}}$$

$$\Rightarrow \Delta G_{\text{mix}} \neq 0$$

Incorrect answer, is $\Delta G_{\text{mix}} = 0$

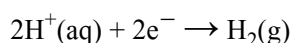
55. Ans (1)

$$Q = ne$$

$$\text{i.t. } n.e$$

$$n = \frac{1 \times 60}{1.6 \times 10^{-19}} = 3.75 \times 10^{20} \text{ electrons}$$

56. Ans (1)



$$\therefore E = E^0 - \frac{0.0591}{2} \log \frac{P_{\text{H}_2}}{[\text{H}^+]^2}$$

$$0 = 0 - 0.0295 \log \frac{P_{\text{H}_2}}{(10^{-7})^2}$$

$$\frac{P_{\text{H}_2}}{(10^{-7})^2} = 1$$

$$P_{\text{H}_2} = 10^{-14} \text{ atm}$$

57. Ans (2)



$$\text{moles of } \text{MgCO}_3 = \frac{20}{84} = 0.238 \text{ mol}$$

From above equation

1 mole MgCO_3 gives 1 mole MgO

$\therefore$ 0.238 mole MgCO_3 will give 0.238 mole

$$\text{MgO} = 0.238 \times 40 \text{ g} = 9.523 \text{ g MgO}$$

Practical yield of $\text{MgO} = 8 \text{ g MgO}$

$$\therefore \% \text{ purity} = \frac{8}{9.523} \times 100 = 84\%$$

58. Ans (2)

Arrhenius equation

$$K = A \cdot e^{-E_a/RT} \Rightarrow \ln K = \ln A - \frac{E_a}{RT}$$

so, activation energy of reaction can be determined

from the slope of $\ln K$ vs $\frac{1}{T}$

59. Ans (1)

$$\frac{w_{H_2}}{w_{O_2}} = \frac{1}{4} \Rightarrow \frac{n_{H_2}}{n_{O_2}} = \frac{1/2}{4/32} = \frac{4}{1}$$

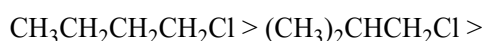
60. Ans (4)

Match List-I with List-II :

	Quantum numbers		Orbital
(a)	$n = 2, \ell = 1$	(i)	2 p
(b)	$n = 3, \ell = 2$	(ii)	3 d
(c)	$n = 3, \ell = 0$	(iii)	3 s
(d)	$n = 2, \ell = 0$	(iv)	2 s

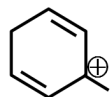
(a)- (iii), (b)-(iv), c-(ii), (d)-(i)

61. Ans (3)



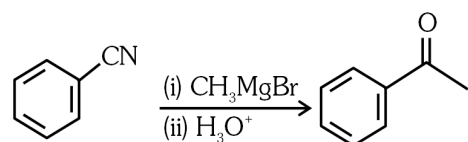
$$\text{Rate of } S_N2 \propto \frac{1}{\text{Steric Hindrance}}$$

62. Ans (4)



3° Cation + resonance stabilized.

63. Ans (4)

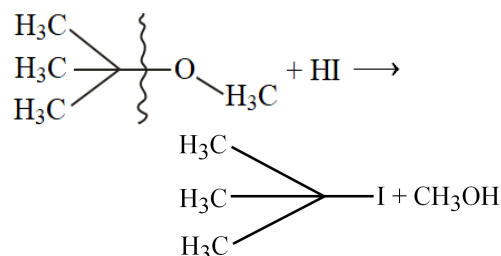


64. Ans (1)

Telegram: @NEETxNOAH

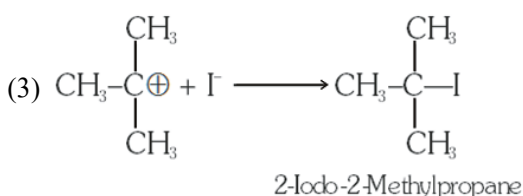
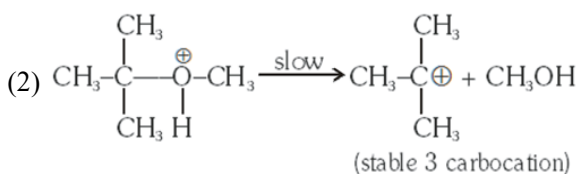
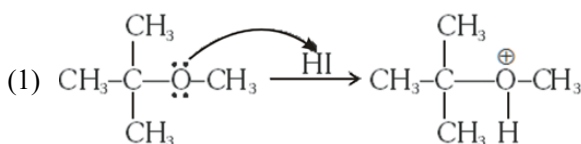
	Amines	pK _b
(a)		3.27
(b)		4.75
(c)		9.30
(d)		9.38

65. Ans (2)

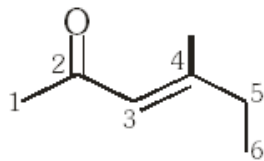


Whenever one of the alkyl group is a tertiary group, the halide formed is a tertiary halide.

Mechanism :



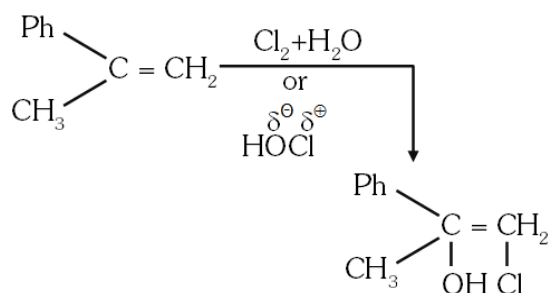
66. Ans (1)



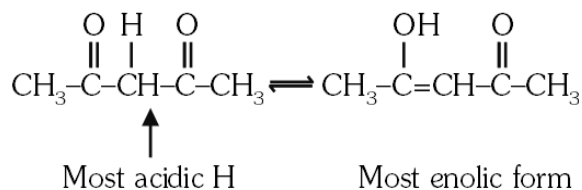
Longest parent chain is selected by giving lowest numbering to ketone

IUPAC name – 4-methyl hex-3-en-2-one

67. Ans (1)

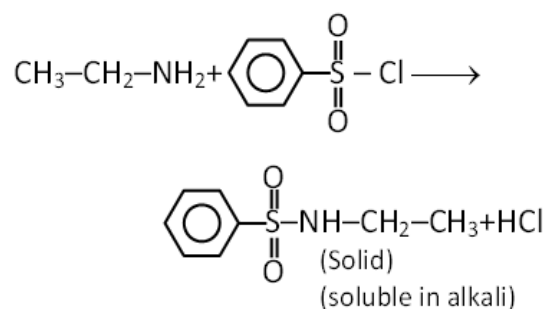


68. Ans (2)

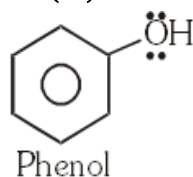


69. Ans (3)

1° amines react with Hinsberg's reagent to give a solid, which dissolves in alkali.



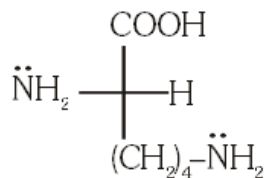
70. Ans (4)



+ R effect of -OH group enhances the reactivity in aromatic electrophilic substitution reaction.

71. Ans (1)

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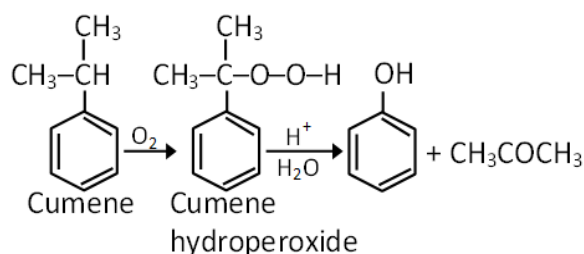


Lysine

Since it contains more number of -NH₂ groups as compared to -COOH groups hence it is basic amino acid.

72. Ans (2)

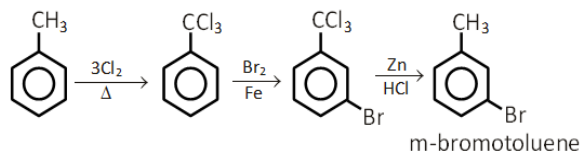
Phenol is manufactured from the hydrocarbon, cumene. Cumene (isopropylbenzene) is oxidised in the presence of air to cumene hydroperoxide. It is converted to phenol and acetone by treating it with dilute acid. Acetone, a by-product of this reaction, is also obtained in large quantities by this method.



73. Ans (1)

Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution because the inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions.

74. Ans (1)



75. Ans (3)

The ortho and para isomers can be separated by steam distillation o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding while p-nitrophenol is less volatile due to intermolecular hydrogen bonding which cause association of molecule.

78. Ans (4)

Dipole - induced dipole occurs between polar & Non-polar molecule

HCl	He
Polar	Non-polar
$\mu \neq 0$	$\mu = 0$

79. Ans (3)

$O_2^- \rightarrow 17 e^-$ contains one unpaired e^- in ABMO.

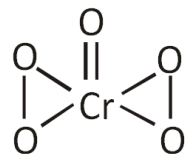
$$s1s^2 s^*1s^2 s2s^2 s^*2s^2 2\sigma 2p_z^2 \pi 2p_x^2 = \pi 2p_y^2$$

$$\pi * 2p_x^2 = \pi * 2p_y^1$$

80. Ans (1)

On moving down the group bond length increases so liberation tendency of H will be more.

81. Ans (3)

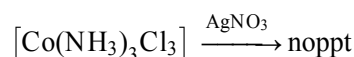
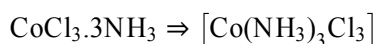


CrO_5 has 2 peroxy linkage.

82. Ans (2)

Due to d^5 configuration and H_2O is a weak ligand.

83. Ans (4)



84. Ans (2)

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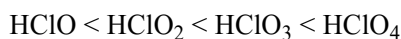
Hexacyanidoferrate (III) ion

85. Ans (1)



86. Ans (2)

Acidic strength $\propto$ EN $\propto$ +ve O.S.



+1 +3 +5 +7

87. Ans (2)

(2) $B < C < N < O$ (given I.P. order)

$B < C < O < N$ (correct)

(3) $I < Br < Cl < F$ (given ΔH_{eg} order)

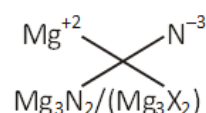
$I < Br < F < Cl$ (Correct)

88. Ans (4)

Magnesium ion = Mg^{+2}

X = Nitrogen

Nitrogen ion = N^{-3}



89. Ans (2)

For same shell

$$[s^1 < p^1 < s^2 < p^2 < p^4 < p^3 < p^5 < p^6]$$

$Li < B < Be < C < O < N < F < Ne$

90. Ans (1)

PCl_5 - Trigonal bipyramidal.

91. Ans (2)

NCERT-XII, Pg. # 60

93. Ans (3)

NCERT-XII, Pg. # 56, 58, Module

95. Ans (1)

NCERT-XII, Pg. # 92

96. Ans (4)

Translation begins when **the small subunit of the ribosome encounters mRNA** and binds to it. This forms the initiation complex, which is followed by the joining of the large subunit to complete the ribosome assembly for translation.

97. Ans (3)

NCERT XII Pg # 99 (E)

NCERT XII Pg # 109, 110 (H)

103. Ans (3)

NCERT-XII, Pg. # 195

104. Ans (4)

NCERT XII Pg # 236 & 237 (E)

NCERT XII Pg # 257, 258 (H)

108. Ans (1)

The correct answer is: 1. Inverted pyramid of biomass

Explanation: In an inverted pyramid of biomass, the biomass of higher trophic levels (such as secondary consumers) is greater than that of the lower trophic levels (such as primary producers).

Here's why:

- Secondary consumer: 120 g
- Primary consumer: 60 g
- Primary producer: 10 g

In this scenario, the biomass at the secondary consumer level (120 g) is greater than that of the primary consumers (60 g) and primary producers (10 g). This results in an inverted pyramid of biomass, where the producers, usually at the base of the pyramid, have a smaller biomass than the consumers above them.

This is an uncommon situation but can occur in certain ecosystems, such as aquatic ecosystems, where the biomass of producers may be smaller than that of consumers due to rapid turnover rates of phytoplankton and other primary producers.

109. Ans (1)

NCERT-XII, Pg. # 219,220

110. Ans (2)

NCERT-XII, Pg. # 224

113. Ans (2)

NCERT-XI, Pg. # 16

114. Ans (1)

NCERT XI Pg # 38

115. Ans (1)

1. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

Explanation:

(a) Spirogyra: It exhibits a dominant haploid free-living gametophyte. In Spirogyra, the gametophyte is dominant, and the sporophyte is short-lived. So, it matches with (ii).

(b) Pteridium: This is a vascular plant (fern) with a dominant diploid sporophyte, and the gametophyte (prothallus) is highly reduced. So, it matches with (iii).

(c) Funaria: This is a moss, which exhibits a dominant haploid leafy gametophyte with a partially dependent sporophyte. So, it matches with (iv).

(d) Cycas: This is a gymnosperm, which has a dominant diploid sporophyte with highly reduced male or female gametophytes. So, it matches with (i).

121. Ans (1)

The correct answer is: 1. Green sulphur bacteria

Explanation: Green sulphur bacteria perform anoxygenic photosynthesis, meaning they do not produce oxygen during the process. Instead, they use substances like hydrogen sulfide (H_2S) as an electron donor, and in the process, sulfur is produced, not oxygen.

In contrast:

- **Nostoc** (a cyanobacterium) performs oxygenic photosynthesis, producing oxygen.
- **Cycas** and **Chara** (both plants) also perform oxygenic photosynthesis, producing oxygen.

125. Ans (4)

NCERT XI Pg # 229

127. Ans (1)

NCERT-XI, Pg. # 63-66

129. Ans (2)

Offsets are a type of vegetative reproduction that occurs in some plants. They are small, lateral shoots that arise from the base of the parent plant, typically forming new plants that are genetically identical to the parent. This process is a result of mitotic divisions, where the cells divide to produce genetically identical daughter cells, leading to the formation of the offset.

131. Ans (1)

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NCERT XI Pg # 96 & 97

132. Ans (1)

The correct matching is:

- (a) Cells with active cell division capacity → (ii) Meristematic tissue
- (b) Tissue having all cells similar in structure and function → (iv) Simple tissue
- (c) Tissue having different types of cells → (i) Vascular tissues
- (d) Dead cells with highly thickened walls and narrow lumen → (iii) Sclereids

So, the correct answer is Option 1. A - II, B - IV, C - I, D - III.

133. Ans (4)

NCERT-XII, Pg. # 13

136. Ans (2)

the answer is a-(ii), b-(iii), c-(iv), d-(i)

137. Ans (2)

NCERT-XI, Pg. # 47, 48

138. Ans (3)

- Gregarious, polyphagous pest - Locusta
- Adult with radial symmetry and larva with bilateral symmetry - Asterias
- Book lungs - Scorpion
- Bioluminescence - Ctenoplane

So, the correct option is (3) (iv) (i) (ii) (iii).

139. Ans (2)

NCERT XI Pg #114

140. Ans (3)

Old NCERT XI Page # 103

141. Ans (2)

NCERT XI Pg # 279 (E)

NCERT XI Pg # 280, 281 (H)

142. Ans (3)

NCERT XI, Page # 194,196

146. Ans (2)

NCERT XI Pg # 334 (E)

NCERT XI Pg # 336 (H)

148. Ans (3)

Hormone - Disease

- **Insulin - Diabetes mellitus** (Insulin deficiency leads to inability to regulate blood sugar levels.)
- **Thyroxin - Goitre** (Thyroxin deficiency can lead to an enlarged thyroid gland, known as goitre.)
- **Corticoids - Addison's disease** (Corticoid deficiency is associated with Addison's disease.)
- **Growth Hormone - Acromegaly** (Excess growth hormone production in adults can cause acromegaly, leading to enlarged bones and tissues.)

Therefore, the correct option is 3:

(a) (V) (b) (IV) (c) (I) (d) (III)

150. Ans (3)

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NCERT XI, Page # 234

152. Ans (1)

NCERT-XI, Pg.# 242, 247

156. Ans (3)

(b) Differentiation of gamete occurs after the completion of meiosis: In **spermatogenesis**, differentiation of sperm occurs after meiosis (i.e., during spermiogenesis), which is true. However, in **oogenesis**, differentiation occurs before meiosis is completed (i.e., during the early stages of oogenesis). Therefore, this statement is **true for spermatogenesis but not oogenesis**.

(c) Meiosis occurs continuously in a mitotically dividing stem cell population: In **spermatogenesis**, meiosis occurs continuously in the testis, as spermatogonia divide mitotically throughout life, and some of them undergo meiosis. In **oogenesis**, meiosis is arrested at specific stages (prophase I until puberty, and metaphase II until fertilization), so this statement is **true for spermatogenesis but not oogenesis**. **(e) It is initiated at puberty:** Spermatogenesis begins at puberty and continues throughout a male's life, whereas **oogenesis** begins during fetal development and is arrested at certain stages until puberty. So, this statement is **true for spermatogenesis but not oogenesis**.

157. Ans (4)

NCERT XII Pg # 46

159. Ans (4)

Cell-mediated immune response is the primary mechanism responsible for the rejection of kidney grafts (or any organ transplant). In this process, **T-cells** recognize the transplanted tissue as foreign and attack it. The immune system identifies the graft as having different human leukocyte antigens (HLAs), which triggers the T-cells to mount an immune response, leading to rejection.

161. Ans (2)

NCERT-XII, Page # 137

162. Ans (2)

SOLUTION:- Karl Ernst von Baer is known for disapproving the idea of **ontogeny recapitulating phylogeny**, which was proposed by Ernst Haeckel. Von Baer argued that embryonic development does not necessarily reflect the evolutionary history of an organism. He stated that the early stages of development are more conserved and do not follow the exact sequence of evolutionary ancestors, which contradicted Haeckel's hypothesis.

163. Ans (4)

NCERT-XII, Pg. # 124, 125

164. Ans (3)

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NCERT-XII, Pg. # 164

165. Ans (1)

NCERT XII Pg # 212

167. Ans (1)

Problem statement : Asking About :
PCR and selection of transformants.

Concept :

PCR - Amplifies the DNA.

Selectable markers used for selection of r-DNA.

Solution / Explanation :

PCR used for DNA fragment amplification.

Ampicillin resistant gene is used for selection of recombinants.

168. Ans (3)

NCERT XI, Page # 181, 182

170. Ans (3)

The correct answer is: 3. Genetic code is nearly universal

Explanation:

The nearly universal nature of the genetic code means that the same codons (sequences of three nucleotides) generally specify the same amino acids across different species. This allows bacteria, which normally do not produce human proteins, to produce human insulin when the corresponding human gene is inserted into their DNA. The bacterial machinery can read the human gene's codons and produce the insulin protein accordingly.

172. **Ans (2)**

NCERT-XI, Pg. # 123

173. **Ans (4)**

NCERT-XI, Pg. # 121

174. **Ans (3)**

NCERT XI Pg # 132

177. **Ans (2)**

NCERT-XI, Pg. # 106, 107

178. **Ans (2)**

NCERT XI Pg # 110

179. **Ans (1)**

Telegram: @NEETxNOAH

Earthworms do not undergo metamorphosis.

Tunicates undergo metamorphosis

Moths undergo complete metamorphosis

Starfish undergo metamorphosis Therefore, the

correct answer is Option 1

180. **Ans (2)**

NCERT XI Pg # 101 & 103